Training Sentence Segmenters on Medieval Languages

Workshop LLcD 25 : Developing Models for Linguistic Research: Transcription and Training Data Usage in Low-Resource Scenarios

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1. Introduction

Aquilign: Project Overview

A multilingual collation project at the intersection of philology and computational methods

- **Automatic segmentation** and **alignment** of **medieval texts** in **multiple languages**
- A **philologically grounded perspective**: textual variants, transmission, and manuscript traditions
- **Methodological focus today**: **automatic segmentation** as a foundation for further NLP processing

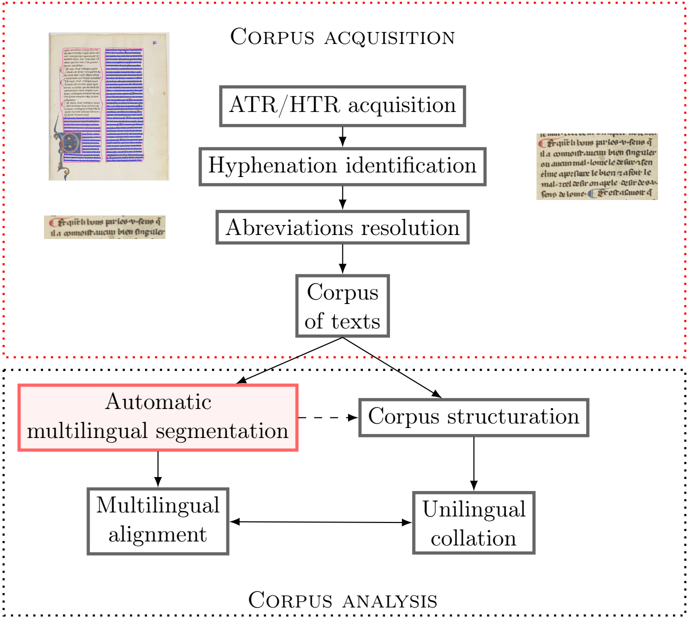
How Semantic Methods Improve Multilingual Alignment

- **Formal alignment fails** on multilingual corpora made of translated texts
- **Semantic representations** are needed for multilingual alignment
- We use **LaBSE**, a multilingual sentence embedding model
- **Bertalign** aligns segments with flexible $n \Leftrightarrow n$ mappings (Liu and Zhu 2023)
- **Ongoing improvement** through fine-tuning on a **domain-specific corpus** (medieval languages)
- The **segmentation task** is the **first step** in the global alignment process : it produces what is going to be aligned

2. Approach

2.1 Global Approach

From Manuscript to Machine-Readable Corpus



Multilingual Alignment Example

Latin (Rome W)	**French**	**Catalan**	**Castilian**	**English**	**Beinecke**
Oportet ergo esse rationalem potentiam, in qua **potest esse virtus**.	Il conuient doncques que la puissance en quoy **uertuz est mise** soit racionele.	Coue donques que la potencia en la qual **ha esser la uirtut** sia racional.	E por ende conuiene que sea poderio razonable aquel en que **esta la uirtud**.	thanne **the vertue moot be** in the resonable vertue and potencial myght.	Oportet ergo esse rationalem potentiam in qua **ponitur esse uirtus**

Table 1: Segment aligned across six versions (focus on the translation of *virtus* in Giles of Rome's *De Regimine Principum*, ca. 1280).

☛ French and Castilian follow the reading *ponitur*, as found in the manuscripts and Zainer's edition, while Catalan and English introduce modal constructions.

2.2 Identifying the Segments to Align

From Contemporary Tools to Medieval Challenges

Recent segmentation tools (e.g., Liu and Zhu 2023) rely on the idea that texts are divided into **well-defined sentences**, thanks to modern punctuation.

However, in medieval texts:

- **Punctuation exists**, but its usage is **unstable** and **not standardized** :
  - it does not necessarily indicate the end of a syntactic unit,
  - **Initial capitalization** is not a reliable clue either.

⚠ The modern notion of a **sentence** is therefore **not suitable** for automatic segmentation of medieval texts.

Absence of sentence in the medieval witnesses

Transcription of a same passage in two different witnesses :

BnF fr. 111 *: ains sem part si tost quil leut commandee a dieu et cheuauche en telle maniere iucqua tierce tant quil uient a lissue de la fourest. et il tourne a destre vers ung chemin uieil et ancien.*

BnF fr. 751 *: Ains sen part si tost comme il lot comandee a dieu. et cheuauche en tel maniere. tant quil uient a lissue de la forest et il torne a destre uers le chemin uiez et ancien*

Detecting Segment Boundaries with BERT

The segmentation model uses a machine learning approach (BERT) to automatically predict where syntactic segments begin.

Each token receives a label:

- **0** = inside a segment
- **1** = beginning of a new segment

un	chemin	viés	et	ancien	si	ne	demora	gueres
0	0	0	0	0	1	0	0	0

Table 2: Example of model output: “si” marks the beginning of a new segment.

See the character based tokenizer by Clérice 2020

2.3 Training Dataset

A Multilingual Corpus for Training Automatic Segmentation of Medieval Texts

We compiled a **rich**, **multilingual**, **manually annotated** reference corpus to support automatic segmentation of medieval texts.

Objectives

- Detect sentence and segment boundaries in non-standardized historical texts.
- Provide training data for automatic segmentation models

Why it matters:

- **Downstream NLP:** parsing, translation, and alignment
- **Access & reuse:** enhancing the accessibility and reusability of medieval sources
- **Scholarly impact:** enabling cross-linguistic comparison and advancing philological and historical-linguistic research

Training Dataset — Overview

- **Period & genres:** 13^{th}–16^{th} c.; narrative, didactic, legal, theological, and scholarly prose, among other genres
- **Sources:** critical editions, specialized corpora, original transcriptions
- **Languages:** Latin, French, Castilian, Portuguese, Catalan, Italian, English
- **Preprocessing:** structural cleanup (punctuation, whitespace)
- **Segmentation:** manual boundaries marked with **£**; non-modernizing; no POS/parse annotations
- **Size:** $\approx$ 70,000 annotated segments, 500.000 tokens
- **Use case:** preprocessing for multilingual alignment (Aquilign)

Processing Pipeline

Get raw texts

(TXT, XML, HTML, PDF)

↓

Clean & normalize

spacing, markup; no orthographic normalization

↓

Annotate

annotate segment boundaries (£)

↓

Store segmented data

Fully segmented excerpts; one file per language

↓

Split into train/dev/test

Partition the segmented data into ML subsets

3. What We Tested So Far

3.1 Rule-based

Rule-based delimiters

Portuguese (pt)

```
"pt": {
  "punctuation": [ "(", ")", ".", ";",
  ↪  ",", "/", "-", "!", ":", "?", "¶"
  ↪  ],
  "word_delimiters": [
    "\\b[Aa]inda que\\b",
    "\\b[Aa]nte que\\b",
    "\\b[Aa] saber\\b",
    "\\b[Bb][eE]n? ass?[yi] como\\b",
    "\\b[aA]taa?\\b",
   "\\b[cC]a[ar]?\\b",
    "... etc ..."
  ]
}
```

Latin (la)

```
"la": {
  "punctuation": [ "(", ")", ".", ";",
  ↪  ",", "/", "!", ":", "?", "¶" ],
  "word_delimiters": [
    "\\b[Aa] quo\\b",
    "\\b[Aa]d quae\\b",
    "\\b[Dd]e quo\\b",
    "\\b[Ee][io] quod\\b",
    "\\b[Cc]u[ij]us\\b",
     "\\b[Cc]um\\b",
    "... etc ..."
  ]
}
```

1: Rule-based delimiters for Portuguese (pt) and Latin (la) — excerpts.

Full rules available at: `https://github.com/ProMeText/multilingual-segmentation-dataset/notes/annotation_guidelines`

Heuristics: from config to segmentation

- **Configuration (JSON):** punctuation + regex patterns per language
- **Segmentation script (utils.py):** loads JSON, applies rules
- **Patterns:**
  - Simple patterns: `\bque\b`, `\bcomo\b`
  - Complex expressions: `\b[tT]anto que\b`
  - Concatenation: `\bde\b\s+\bmodo\b\s+\bque\b`
  - Graphical Variation: `\b[Pp]er ?ci[òo] ?ch[eé]?\b`
- **Outcome:** boundaries inserted as `£`

3.2 Learning-based

SaT : Segment any Text

Segment Any Text, Frohmann et al. 2024 :
`https://github.com/segment-any-text/wtpsplit`

- designed for **85 contemporary languages**, based on BERT AutoModelForTokenClassification models
- to resolve the problem of the sentence (not clean data from the Web, for instance)
- segments **raw text into sentences** (full sentences from the point view of what is a modern sentence) not in phrases

Example on medieval text : *['et tant qu'il brisent les mireoirs as piez tant vont entour lors ne truevent riens illec et ainsi s'en eschapent ceuls qui là sont', 'et aucunes foiz est avenu de ces bestes que eles pensent tant à leur figures remirer et en sont aucunes']*
→ only two sentences have been identified

Architectures

- → We first used a BERT-Based Segmenter (Token Classification): `AutoModelForTokenClassification` (`bert-base-multilingual-cased`): around 177M parameters, 700Mb
- **Is it the simplest solution ? Can we simplify and use lighter architectures ?**
- Different architectures:
    - **Convolutional Neural Networks (CNN)** that perform well on text segmentation tasks (Clérice 2020)
    - **LSTM network**
- Multiple hyperparameters:
    - Language embeddings
    - Attention layer
    - Number of units, hidden size
    - Tokenizer ad embeddings: BERT or homemade

3.3 Results

Results comparison

Model	Tokenizer & Embeddings	Precision	Recall	F1-score	Parameters
Regexp engine	Regular expressions	0.63	0.55	0.59	None
Bert Model	Bert	**0.897**	**0.900**	**0.899**	1.77e-09
DistilBert Model	DistilBert	**0.894**	**0.902**	**0.898**	1.34e-09
CNN Model	Bert Embeddings (pretrained)	0.827	0.814	0.820	1.02e-08
	Custom Tokenizer	0.835	0.830	0.833	2.57e-07
LSTM Model	Bert Embeddings (pretrained)	0.840	0.846	0.843	9.36e-07
	Custom Tokenizer	0.830	0.841	0.835	2.40-e07

Table 3: Comparison of models and tokenization strategies. Results on test data (in-domain)

- **Best hyperparameters** are found with a specific algorithm (Optuna, Akiba et al. 2019)
- Except for BERT models (not tested yet), results are improved when lang metadata embeddings are concatenated to the word embeddings

Evaluation by Language on Best Model (BERT)

Language	Precision	Recall	F1-score
Portuguese	0.968	0.957	0.963
Catalan	0.943	0.931	0.937
French	0.917	0.933	0.925
Italian	0.897	0.908	0.903
Castilian	0.875	0.855	0.865
Latin	0.849	0.864	0.856
English	0.819	0.83	0.824

Table 4: Per-language evaluation results.

Results vary: around 90% and more for Portuguese, French, Italian, Catalan; English has the poorest results (less data, fluctuation in annotation). Very high results for Portuguese and Catalan: annotation that follows more the form.

Ambiguous delimiters: coordinating conjunction *and* across languages

- Some tokens are **ambiguous**: they can be **delimiters**, or not
  - ☛ In our segmentation rules, the coordinating conjunction ("et" in French, for instance) is a good example of this ambiguity:
    puis que je suis en court de droit et de justice, et que je voy mon ennemy devant moy
    → first *et* not delimiter, second *et* delimiter
- Rule: delimiter when it coordinates **two sentences, phrases** or more than two elements. Otherwise, when only two elements → not a delimiter.
- Therefore, the model has to learn to discriminate. The regexp model can't deal with this ambiguity.

One case of ambiguity: the coordinating conjunction

	As Segment Content			As Segment Boundary		
	Precision	Recall	F1	Precision	Recall	F1
English	0.686	0.800	0.738	0.910	0.847	0.878
Catalan	0.760	0.934	0.838	0.951	0.813	0.876
French	0.818	0.778	0.797	0.844	0.874	0.859
Latin	0.695	0.838	0.760	0.890	0.780	0.832
Italian	0.660	0.782	0.716	0.851	0.757	0.801
Portuguese	0.788	0.700	0.741	0.752	0.829	0.789
Castilian	0.889	0.460	0.606	0.495	0.901	0.639

Table 5: Ambiguity management of the LSTM best model

4. Conclusion

4.1 Concluding remarks

Conclusion

- **Identifying effective architectures:** BERT and DistilBert models perform better than simpler CNN and LSTM models. LSTM models seem to outperform CNN architectures on this task.
- **Data availability matters:** Increasing the amount of training data is expected to significantly improve performance, especially given the variability across medieval languages.

One question remains : the **segmentation task** performs **well**, but does it have an incidence on the **main task of alignment** ?
→ evaluation of the **alignment results** from this segmentation has to be done

4.2 Future Directions

Next Steps

Segmentation pipeline

- Expansion of the dataset
- Annotations cross-reviewed among team members
- Focus on underrepresented languages (to improve model performance), add "real life data" (HTR)
- Test the ambiguity management of the models

Overall project

- Improvement of the alignment phase by training LaBSE models, which includes the production of parallel data (we already have a corpus of Bible translations).

Thank you for your attention!

- **Code**: `https://github.com/prometext/aquilign`
- **Multilingual BERT model**: `https://huggingface.co/ProMeText/aquilign-multilingual-segmenter`
- **Segmented dataset**: `https://github.com/ProMeText/multilingual-segmentation-dataset`

Associated references:

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